PH7002: Concepts in Statistical Mechanics
§5 Transfer Matrix

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Reading

Yeomans Chapter 5.

Most of this chapter will be concerned with the **transfer matrix method**, first introduced by **Kramers**, which reduces the problem of calculating the partition function to the problem of finding the eigenvalues of a certain matrix.

Important themes

1. Write partition function in terms of transfer matrix.
2. Thermodynamic properties related to largest eigenvalue of transfer matrix.
3. Perron-Frobenius theorem.
4. Exact result of 1-dim Ising model.

1D Ising model (ad hoc)

We start with the Ising model Hamiltonian $\mathcal{H}$ with nearest neighbor interactions in one dimension:

Define

Then the partition function for N spins is

First we solve the case with no constraints on the boundary spins, and with the external field $h = 0$. Then

Define a new variable

so that

To specify completely the state of the system, we need to provide the set of numbers $\{S_1, \eta_1, \ldots, \eta_{N-1}\}$. Hence we can write the partition function as

The first factor of 2 comes from the sum over S_i and the $N-1$ factors of $2\cosh K$ arise from factorizing the sums over the η variables.

Now we solve the same problem, but with periodic boundary conditions:

The partition function is now

In this case, the state of the system is still specified by the variables

In terms of these variables, we can write

where we have used the fact that $S_i^2 = 1$.

The technical difficulty arises from implementing the periodic boundary conditions properly. In this case, we find that

Another method for $h = 0$ is to use recursion. The partition function for a chain of N spins is

where we have used free, not periodic, boundary conditions. From this we work out the partition function for a chain with $N + 1$ spins:

The sum

Now do the remaining sums to obtain

Iterating this recursion relation gives

i.e., $Z(N) = Z(1)(2\cosh K)^{N-1}$. But

Thus,

Now that we have calculated the partition function in zero field using two different boundary conditions, let us see how they affect the thermodynamics. The free energy, calculated for the case with free boundary conditions is

As $N \to \infty$ (thermodynamic limit),

which is what we would obtain from using periodic boundary conditions. As expected, the difference between boundary conditions becomes negligible as the system size grows.

The transfer matrix

We start with the case of periodic boundary conditions: $S_{N+1} = S_1$.

Now we factorize the summand as follows:

We can think of each term as being like a matrix element of a matrix $\mathbf{T}$

Observe that S_1 and S_2 are the labels of the matrix elements:

Then

To see what this means, recall that the rule of matrix multiplication is that the matrix elements are given by

Also, the trace of a matrix is defined as

Then we see that performing the sums is just matrix multiplication. Thus

Q: *How do we compute* $Tr(\mathbf{T}^N)$?

The factor $\mathbf{T}^N$ is at first sight rather forbidding, but we can readily diagonalize $\mathbf{T}$, by multiplying and post-multiplying with a matrix $\mathbf{S}$ whose rows and columns are eigenvectors of $\mathbf{T}$, i.e. perform a similarity transformation. Since $\mathbf{T}$ is real and symmetric, then $\mathbf{S}^T = \mathbf{S}^{-1}$ and the diagonalized form of $\mathbf{T}$ is

where λ_1 and λ_2 are the eigenvalues of $\mathbf{T}$. This gives

Now use the cyclic property of the trace operation, which implies that

Now consider the case that $\lambda_1 \neq \lambda_2$. Assuming that $\lambda_1 > \lambda_2$ we have

and in the thermodynamic limit $N \to \infty$

where $\alpha = \log(\lambda_1/\lambda_2)$ is a positive constant. So only the largest eigenvalue of the transfer matrix is important in the thermodynamic limit. The free energy is then given by

We can easily compute λ_1. The eigenvalues of $\mathbf{T}$ are given by

Phase transitions

How does a phase transition arise? For $T > 0$, the argument of the square root

for the free energy is positive, for real h and K. This expression for f is manifestly analytic for $T > 0$. More systematically, the only possibilities for the occurrence of a phase transition are: that λ_1 is a non-analytic function of K and h, that the square root vanishes, in which case the eigenvalues become degenerate: $\lambda_1 = \lambda_2$, or that $\lambda_1 = 0$. For $T > 0$ none of these things can happen. We see this here by inspection, but it also follows from **Perron-Frobenius theorem**.

Perron-Frobenius theorem

For an $N \times N$ matrix ($N < \infty$) $\mathbf{A}$ with $A_{ij} > 0$ for all i, j, the eigenvalue of largest magnitude is:

a. real and positive,
b. non-degenerate,
c. an analytic function of A_{ij}.

In one dimension, the transfer matrix for finite-ranged interactions satisfies the requirements of **Perron-Frobenius theorem**. Thus, we immediately find that

and hence that there is no phase transition for $T > 0$. Perron-Frobenius theorem can be used to show the absence of phase transitions at non-zero temperature in **one dimension** for systems with sufficiently **short-ranged interactions**.

In terms of the mathematics, what is special about one dimension: why can the theorem not be applied in higher dimensions to show that phase transitions never occur for $T > 0$? In one dimension the transfer matrix for the Ising model is a 2×2 matrix. A little reflection reveals that in two and higher dimensions, the transfer matrix is a $\infty \times \infty$ matrix in the thermodynamic limit, and so **Perron-Frobenius theorem** does not apply.

At $T = 0$ or equivalently $K = \beta J \to \infty$. The largest eigenvalue of the transfer matrix becomes

Recalling that

Thus,

We do not need to take the thermodynamic limit to obtain non-analytic behaviour at $T = 0$. The magnetization is

Note that the non-analytic behaviour came from a term $\sqrt{h^2} = |h|$. For $T > 0$, this term is of the form $\sqrt{h^2 + \epsilon^2}$ which is analytic at $h = 0$ as long as the constant $\epsilon \neq 0$.

Thermodynamic properties

We start with the case $h = 0$. Then

and $Z = (2\cosh K)^N$ as $N \to \infty$. We have dropped the contribution of λ_2, which in this case is $(2\sinh K)^N$. The free energy is then

The low and high temperature limits

As anticipated, the high temperature limit corresponds to the entropy dominating the free energy, whereas the low temperature behaviour is determined mainly by the energy.

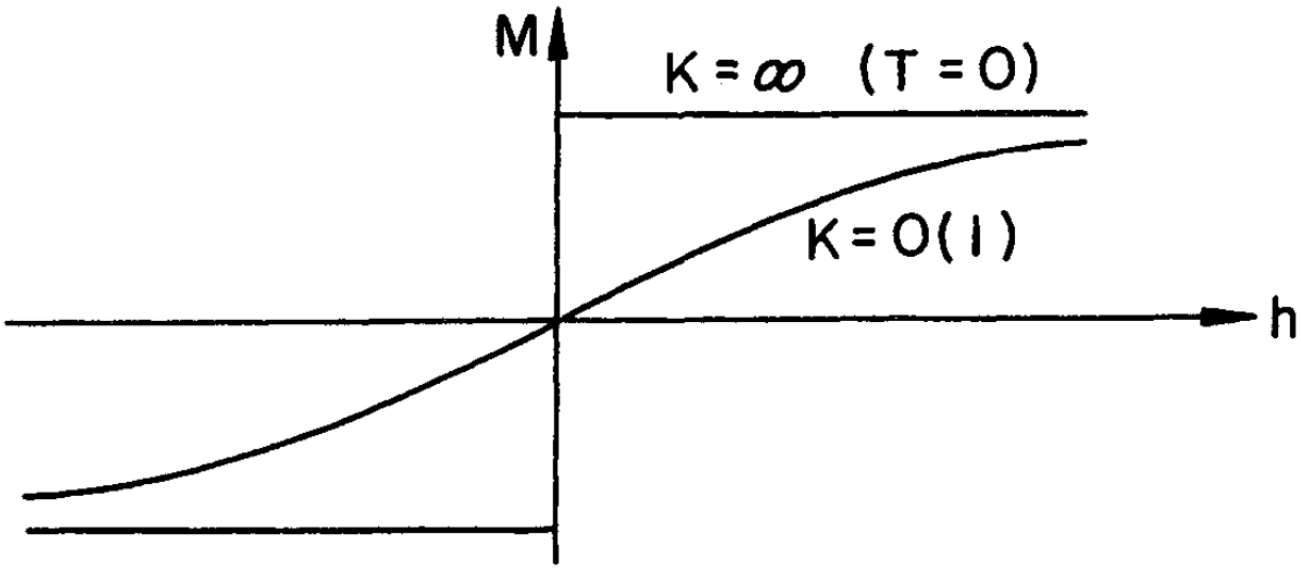
The specific heat is simply obtained from the internal energy E:

Therefore

The heat capacity does not exhibit any singularity, but note the presence of a peak near $J \sim k_B T$, which is sometimes known as a **Schottky anomaly**.

To calculate the magnetization, write the free energy per spin in the form

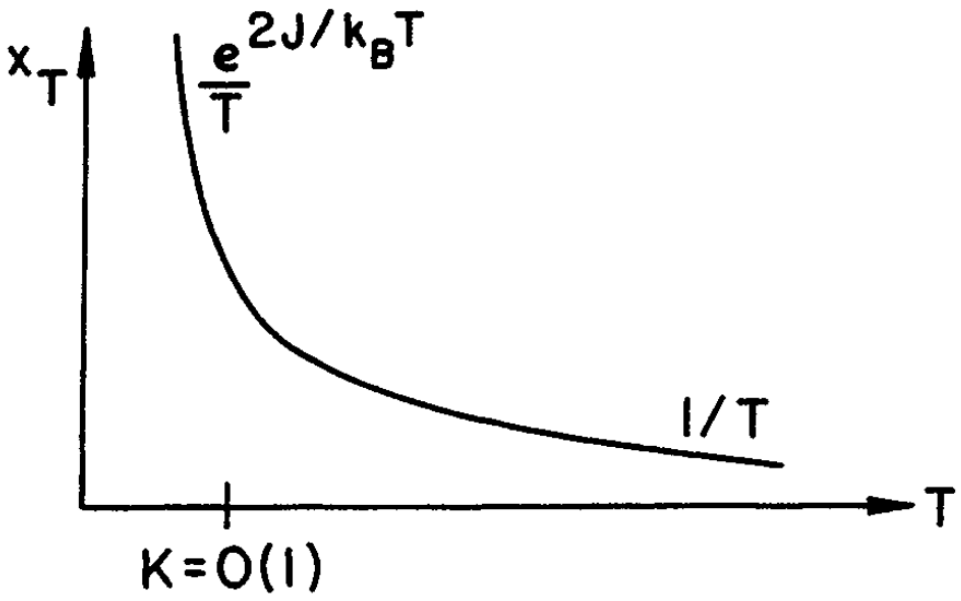
where $w^2 = e^{-4K}$ is the relative probability of the two configurations below, which differ by a single spin flip:


M
K = ∞ (T = 0)
K = O(1)
h

The isothermal magnetic susceptibility χ_T describes how the magnetization changes in response to an external field:

The low and high temperature behaviours are

Note the exponential divergence at zero temperature.



Example 5.1: 1d spin 1 Ising model

Yeomans problem 5.3

5.3 Write down the transfer matrix for the one-dimensional spin-1 Ising model in zero field which is described by the Hamiltonian

$$\mathcal{H} = -J \sum_i s_i s_{i+1}, \quad s_i = \pm 1, 0.$$

Hence calculate the free energy per spin of this model and show that it has the expected behaviour in the limits $T \to 0$ and $T \to \infty$.
[Answer: $f = -kT \ln\{(1+2\cosh\beta J+[(2\cosh\beta J-1)^2+8]^{1/2})/2\}$.]

Spatial correlations

For $T > 0$, we have seen that

The two-point correlation function is defined as

for $h = 0$ and $i < j$. The two-point correlation function may be written in the slightly more transparent form:

showing that G actually measures the correlation in the fluctuation of the spins at different sites.

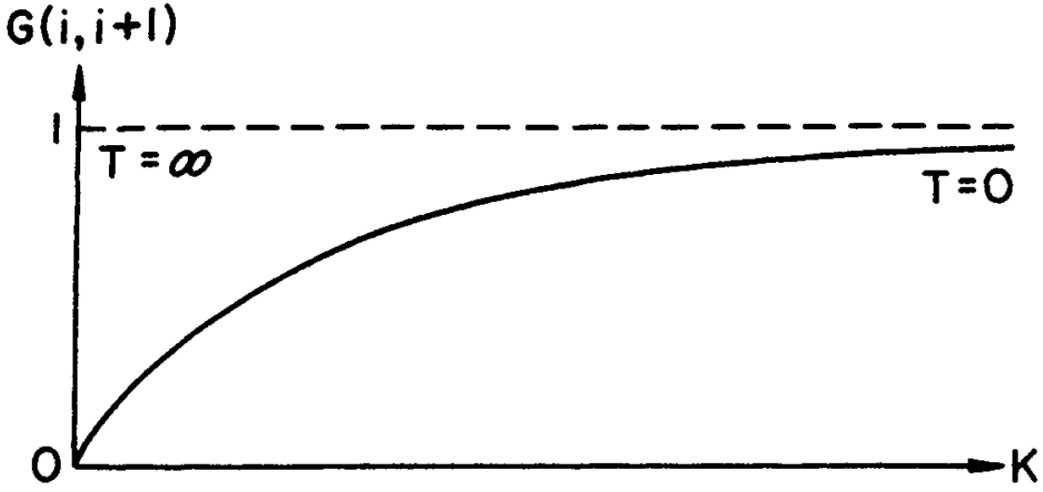
Proceeding to the calculation, we first replace the exchange between spins i and j by J_{ij}, a value that, in principle, varies from bond to bond on the chain. Then

Now we will derive the form of $G(i, j)$ by successive differentiation. There are two steps to the argument. First, we let $j = i + 1$, and calculate $G(i, i + 1)$.

Step 1: In this case,

We can find $Z_N\{K_i\}$ from our previous argument when all the $K_i = K$ - we simply get factors of $\cosh K_i$ instead of $\cosh K$ at each recursion. Thus,

Putting $K_i = K$, we see that neighbors are strongly correlated at low temperature ($K \to \infty$), whereas at high temperature ($K \to 0$), the correlations diminish.



We can derive $\langle S_i S_{i+1} \rangle$ by calculating the energy.

But $\langle S_i S_{i+1} \rangle$ does not depend on i in a translationally invariant state, so

Step 2: How do we go beyond $\langle S_i S_{i+1} \rangle$? Notice that

where we have used the fact that $S_i^2 = 1$. Therefore

Proceeding by induction,

Now set all the $K_i = K$:

The result is translationally invariant as expected when K_i does not depend on i. $G(i, i+j)$ only depends on j. In a continuous system such as a fluid, the analogous statement is that the two-point correlation function satisfies $G(\mathbf{r}, \mathbf{r}') = G(\mathbf{r} - \mathbf{r}')$. This applies even for a finite system, as long as i, j are not near the boundaries, and for a system with periodic boundary conditions.

What do we learn from the two-point correlation function? As expected, there are two possible cases: For $T = 0$ ($K = \infty$), we have $\tanh K = 1$ and

Thus, at zero temperature, the probability that $S_i = S_j$ is unity for all j. This is a perfectly correlated state, and one that exhibits **long range order**. The two states of the system consistent with this are

For $T \neq 0$ ($K \neq \infty$) we have that $\tanh K < 1$ and $\coth K > 1$ $\Rightarrow \log \coth K > 0$. Therefore, for $j > 0$, we can write

showing that correlations decay exponentially for $T > 0$. The correlation length ξ is defined by

where ξ is measured in units of the lattice spacing a.

ξ is a measure of the length over which spins are correlated with probability ~ 1. As $K \to \infty$ i.e. as $T \to 0$, the correlation length diverges to infinity, whereas at high temperature, $\xi \to 0$. How does ξ approach infinity as $T \to 0$? For $K \gg 1$

Therefore,

as $T \to 0$.

There is an essential singularity in the correlation length as the temperature approaches zero, which is the transition temperature in this model: ξ diverges exponentially fast. In fact, we shall generally find that near a continuous transition, the correlation length diverges with some exponent ν, but the correlation length varies according to

and not $\xi \sim \exp[J/k_B(T - Tc)]$, as we have in this one dimensional example.

Correlation functions can also be computed using the transfer matrix formalism. The basic trick is as follows:

Focus on the section of the string of transfer matrices near S_i:

The result of this string is itself a matrix, $\mathbf{A}$, whose ab element is

But this is the same as

The matrix sandwiched between the transfer matrices is one of the Pauli matrices, usually denoted by σ_z. Finally, using the cyclic property of the trace operation,

We can evaluate this expression by using the similarity transformation and the cyclic property of the trace:

We can explicitly compute S: let

be normalized eigenvectors of $\mathbf{T}$ with eigenvalues $\lambda_{1,2}$. Then

Write

so that

using the result that

Note that in the above equations, e, f, g, k etc. are all functions of h, K, which can be explicitly evaluated.

In the limit that $N \to \infty$.

Similarly, the two-point correlation function is

In the limit that $N \to \infty$.

and the correlation function becomes

We can read off the correlation length

and since $\lambda_1 > \lambda_2$ we see again that correlations decay exponentially to zero for $T > 0$. We can also check our previous results for $h = 0$: in that case

Note:

- ξ cannot diverge unless $\lambda_1 = \lambda_2$. There cannot be a phase transition unless the largest eigenvalue of the transfer matrix becomes degenerate. This is a general result.
- For $h \neq 0$, $\lambda_1 > \lambda_2$, so there cannot be a phase transition for $h \neq 0$.
- For $h = 0$, $\lambda_1 = \lambda_2$ when $K = \infty$, so that a phase transition can occur at $T = 0$, as we have already seen.

Change Boundary Conditions

Q: *What happens when we change the boundary conditions?*

Let us see how the partition function changes. In this case, we find that

The Helmholtz free energy is

Example 5.2: q state Pott model

Yeomans problem 5.2

5.2 (i) Write down the transfer matrix for the one-dimensional, q-state Potts model which is described by the Hamiltonian

$$\mathcal{H} = -J \sum_i \delta_{\sigma_i \sigma_{i+1}}, \quad \sigma_i = 1, 2 \ldots q.$$

(ii) Show that the largest eigenvalue is $e^{\beta J} + q - 1$ and that the remaining eigenvalues are all degenerate and take the value $e^{\beta J} - 1$.
(iii) Write down expressions for the free energy and correlation length of the model and show that they take sensible values in the limits of zero and infinite temperature.

Extra Notes